

NAGOYA, Japan

WMO: 476360

Lat: 35.1678N

Lon: 136.9656E

Elev: 56

StdP: 100.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	-0.3	-10.2	1.6	5.2	-9.0	1.8	4.8	9.4	6.0	8.6	6.2	2.6	350	0.412

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	34.9	24.9	33.6	24.7	32.4	24.3	26.1	31.5	25.6	30.8	25.2	30.1	3.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	20.0	27.9	24.3	19.4	27.7	23.9	18.8	27.7	81.1	31.6	79.2	30.9	77.4	30.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	-3.2	37.0	1.2	1.4	-4.0	38.0	-4.7	38.9	-5.4	39.7	-6.2	40.7		
(5)			-4.6	27.0	0.9	0.6	-5.3	27.5	-5.8	27.8	-6.3	28.1	-6.9	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	4.9	5.8	9.3	14.8	19.7	23.3	27.5	28.9	25.0	19.1	12.8	7.4	(6)
(7)		DBStd	8.62	2.13	2.69	3.01	3.14	2.38	2.24	2.51	2.08	2.78	2.76	2.96	2.74	(7)
(8)		HDD10.0	429	159	121	50	3	0	0	0	0	0	0	8	89	(8)
(9)		HDD18.3	1699	417	351	280	110	13	1	0	0	0	23	167	339	(9)
(10)		CDD10.0	2842	0	3	29	148	302	400	542	585	449	283	92	8	(10)
(11)		CDD18.3	1069	0	0	0	6	57	150	283	327	199	48	1	0	(11)
(12)		CDH23.3	9571	0	0	0	31	291	885	2917	3743	1537	167	1	0	(12)
(13)	CDH26.7	3594	0	0	0	2	49	223	1186	1652	468	16	0	0	(13)	
(14)	Wind	WSAvg	3.0	3.1	3.4	3.5	3.3	3.1	2.7	2.8	2.9	2.8	2.7	2.6	2.9	(14)
(15)	Precipitation	PrecAvg	1560	47	64	112	138	155	199	207	146	224	133	78	48	(15)
(16)		PrecMax	2033	155	158	211	230	331	514	450	379	708	530	230	125	(16)
(17)		PrecMin	905	1	9	3	7	51	58	29	4	31	10	7	0	(17)
(18)		PrecStd	258	33	37	47	51	63	80	105	95	129	92	48	30	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	16.8	20.7	26.3	30.2	32.7	36.3	37.1	33.8	28.4	22.4	17.2	(19)
(20)			MCWB	7.7	10.8	12.0	15.2	18.0	23.5	25.0	25.2	24.2	20.7	15.9	12.3	(20)
(21)		2%	DB	11.5	14.1	18.4	24.0	28.0	30.6	34.8	35.4	32.1	26.6	20.7	14.8	(21)
(22)			MCWB	6.6	8.1	11.0	14.7	17.5	21.7	24.8	25.0	23.6	19.3	14.5	9.8	(22)
(23)		5%	DB	10.2	12.3	16.6	22.1	26.4	29.1	33.4	34.2	30.8	25.2	19.2	13.4	(23)
(24)			MCWB	5.7	7.1	10.5	13.9	17.0	21.1	24.6	24.9	22.9	18.7	13.7	8.9	(24)
(25)		10%	DB	9.0	10.7	14.9	20.4	24.8	27.7	32.1	33.0	29.4	23.8	17.7	12.0	(25)
(26)			MCWB	4.8	6.1	9.3	13.5	16.6	20.5	24.4	24.6	22.5	18.0	13.2	7.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	12.2	15.1	19.6	22.0	25.1	26.5	27.0	25.8	23.4	18.1	13.6	(27)
(28)			MCDWB	11.6	15.2	16.5	21.0	24.4	29.7	32.5	33.2	29.8	25.3	19.7	15.0	(28)
(29)		2%	WB	7.7	9.9	13.4	17.5	20.5	24.1	25.9	26.2	25.1	21.7	16.5	11.2	(29)
(30)			MCDWB	10.0	12.6	16.4	20.6	23.5	27.6	31.5	31.8	29.0	24.7	18.6	13.8	(30)
(31)		5%	WB	6.4	8.2	11.9	16.2	19.2	23.2	25.4	25.8	24.5	20.2	15.3	9.6	(31)
(32)			MCDWB	9.2	10.9	15.0	19.9	23.3	26.7	30.7	31.1	28.3	23.4	17.6	12.4	(32)
(33)		10%	WB	5.3	6.6	10.3	15.0	18.2	22.3	25.0	25.4	24.0	19.1	14.1	8.4	(33)
(34)			MCDWB	8.4	9.9	13.9	18.8	22.5	25.7	29.9	30.5	27.7	22.5	16.9	11.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.2	9.1	9.5	8.9	7.4	7.1	7.7	7.2	7.7	7.9	7.4	(35)
(36)			MCDBR	9.0	10.1	11.2	11.8	11.5	9.5	9.1	9.1	8.5	8.8	9.2	8.7	(36)
(37)			MCWBR	5.5	6.2	6.5	5.5	4.7	3.5	3.0	2.8	3.1	4.0	5.1	5.4	(37)
(38)		5% WB	MCDWB	7.8	8.6	8.7	8.4	8.3	6.9	7.8	7.8	6.9	6.9	7.1	7.3	(38)
(39)			MCWBR	5.6	6.3	7.0	5.5	4.9	3.2	2.9	2.8	3.1	4.3	5.2	5.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.388	0.427	0.493	0.536	0.544	0.573	0.562	0.540	0.493	0.456	0.412	0.387	(40)	
(41)		taud	2.181	2.085	1.932	1.878	1.925	1.926	2.008	2.058	2.157	2.179	2.210	2.205	(41)	
(42)		Ebn at Noon	786	797	775	764	764	740	744	753	769	760	754	760	(42)	
(43)		Edh at Noon	114	141	178	198	191	191	175	163	141	127	111	106	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.08	4.02	4.79	5.24	4.72	4.92	5.06	4.02	3.27	2.54	2.12	(44)	
(45)		RadStd	0.17	0.23	0.28	0.46	0.57	0.43	0.69	0.47	0.36	0.32	0.22	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-0.62	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page